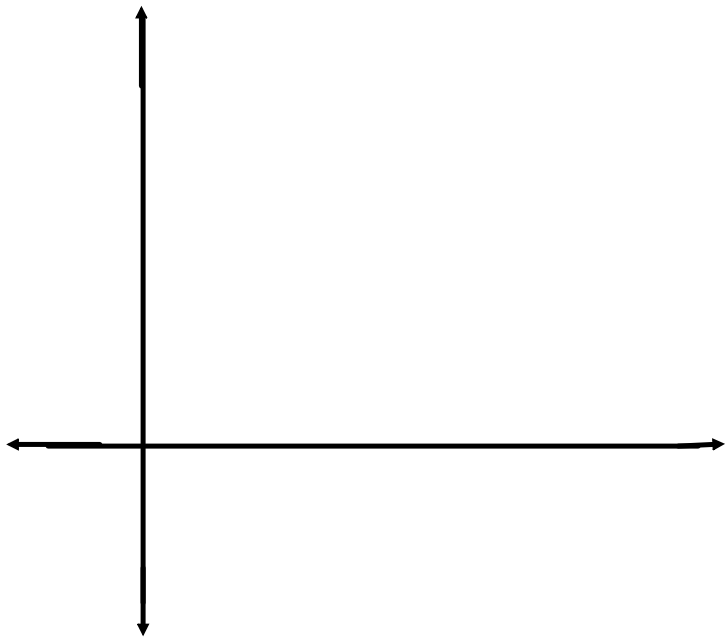
Intermediate Value Theorem

A continuous function $f(x)$ on $[a, b]$ takes on every value between $f(a)$ and $f(b)$.

Or . . . $f(c) = y_0$ must have at least one solution if y_0 is between $f(a)$ and $f(b)$.



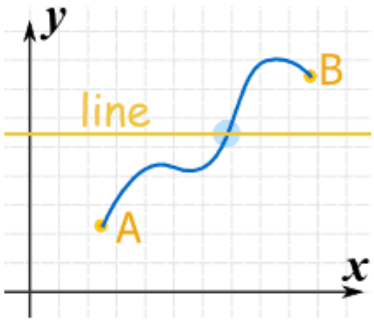
Given: $f(1) = 3$ and $f(4) = -8$ and
$f(x)$ is continuous on $[1, 4]$.

We can guarantee the existence of a solution to the equation $f(x) = -5$.

What about $f(x) = -9$?

Intermediate Value Theorem

The idea behind the Intermediate Value Theorem is this:



When we have **two points** connected by a continuous curve:

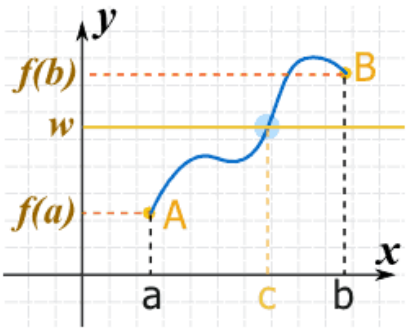
- one point below the line
- the other point above the line

... then there will be **at least one place** where the curve crosses the line!

Well **of course** we must cross the line to get from A to B!

More Formal

Here is that idea stated more formally:



When:

- The curve is the function $y = f(x)$,
- which is continuous on the interval [a, b],
- and w is a number between f(a) and f(b),

Then ...

... there must be at least one value **c** within [a, b] such that **f(c) = w**

In other words the function $y = f(x)$ at some point must be $w = f(c)$

Notice that:

- **w** is between f(a) and f(b), *which leads to ...*
- **c** must be between a and b

Example: is there a solution to $x^5 - 2x^3 - 2 = 0$ between x=0 and x=2?

At x=0: